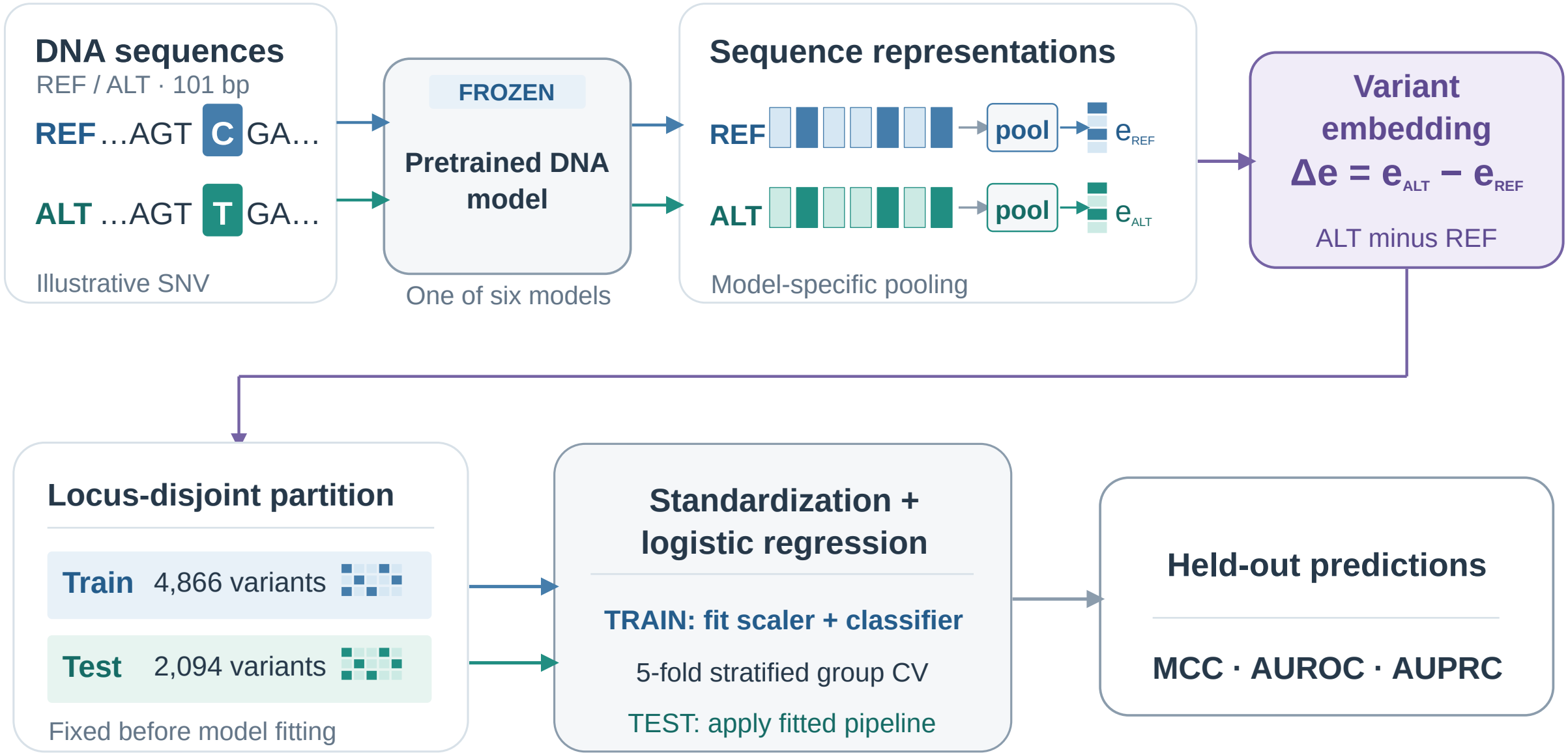

DNA sequences
REF / ALT · 101 bp
REF …AGT C GA…
ALT …AGT T GA…
Illustrative SNV
FROZEN
Pretrained DNA model
One of six models
Sequence representations
REF
pool
e_REF
ALT
pool
e_ALT
Model-specific pooling
Variant embedding
Δe = e_ALT − e_REF
ALT minus REF
Locus-disjoint partition
Train 4,866 variants
Test 2,094 variants
Fixed before model fitting
Standardization + logistic regression
TRAIN: fit scaler + classifier
5-fold stratified group CV
TEST: apply fitted pipeline
Held-out predictions
MCC · AUROC · AUPRC